The non-equilibrium thermodynamics of quantal response:

criticality escape, a supercritical wedge, and the decoupling of response asymmetry from dissipation

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Abstract

Logit revision dynamics on a finite game define a Markov jump process on the joint profile space whose stationary state is a genuine thermodynamic object: a Gibbs equilibrium exactly when the game is potential, and a dissipative non-equilibrium steady state otherwise. We sweep the two-parameter plane spanned by the harmonic fraction α of the game's Hodge decomposition and the logit precision λ, reading four calibrated meters at every point — spectral criticality $\rho(SB)$, reciprocity defect $\mathcal{R}$, exact entropy production σ_{EP}, and its trajectory estimators — and report three facts we did not predict. (i) *Criticality escape*: on potential games $\rho(SB)$ is non-monotone in λ — equilibrium concentration kills the choice covariance faster than λ amplifies it — while harmonic games cross $\rho = 1$ and stay there, opening a *supercritical wedge* whose frontier descends in α (median onset $\alpha = 0.5$). (ii) *Decoupling*: dissipation and response asymmetry co-move almost perfectly along α ($\rho_S = 0.993$ marginally) yet, conditional on α, their correlation degrades and *reverses* to -0.355 in the near-harmonic regime; a mechanism experiment shows this is structural — the equilibrium susceptibility is a local derivative, the entropy production a global flux functional, and no renormalisation of the former recovers the latter's ordering. A repair hypothesis we proposed is refuted by its own preregistered test. (iii) *Estimability, through to real data*: a k-th order Markov KLD estimator tracks the exact meter with rank correlation 1.0; a debiased finite-time thermodynamic-uncertainty bound certifies dissipation from two cumulants of one current; and, pointed at a real power market (CAISO SP15, July 2026), the instruments deliver a verified detection — day-ahead hourly prices violate pair-level detailed balance at ≈ 1.1 nats/day, concentrated in scarcity weeks — alongside an equally instructive certified at-null verdict for the 5-minute series and one documented retraction en route. Every number regenerates from fixed seeds; every claim carries its confidence tier and its falsifier.

1 Setup: the joint-profile chain and its four meters

A finite game with players $i = 1..N$ and payoff tensors u_i defines *Glauber–logit dynamics*: revision opportunities arrive at unit rate per player, and the revising player redraws an action from $\mathrm{softmax}(\lambda\, u_i(\cdot, a_{-i}))$. This is a continuous-time Markov jump process on the joint profile space $\prod_i A_i$. Two exact facts anchor everything below (both are permanent regression tests in the library, verified to 10^{-10}–10^{-16}; tiers per the claims ledger).

*With agent-assisted implementation. Code, gate records, and every number's regeneration path: https://github.com/SharathSPhD/sage.

Potential games are thermal equilibrium. If the game admits an exact potential Φ, the dynamics are reversible with Gibbs stationary law $\pi \propto e^{\lambda\Phi}$ [1]: detailed balance holds, the stationary probability current $J^*(a,a') = \pi(a)w(a\to a') - \pi(a')w(a'\to a)$ vanishes edge-by-edge, and the Schnakenberg entropy production rate [4]

$$\sigma_{\mathrm{EP}} \;=\; \tfrac{1}{2}\sum_{a,a'}\left[\pi(a)w(a\to a') - \pi(a')w(a'\to a)\right]\log\frac{\pi(a)w(a\to a')}{\pi(a')w(a'\to a)} \;\geq 0 \tag{1}$$

is exactly zero.

Harmonic content is the drive. The Hodge decomposition [2] splits any game into potential, harmonic and non-strategic components; we write $\alpha \in [0,1]$ for the harmonic fraction of the normalised game. At $\alpha > 0$ the same dynamics settle into a non-equilibrium steady state: stationary but circulating ($J^* \neq 0$), dissipating at rate $\sigma_{\mathrm{EP}} > 0$.

The meters. At each (α,λ) we read: (1) the *feedback gain* $\rho(SB)$ — spectral radius of the strategic feedback operator in $\chi^{\mathrm{eq}} = (I - SB)^{-1}S$, with $\rho \geq 1$ marking a near-singular resolvent (criticality); (2) the *reciprocity defect* $\mathcal{R} = \|\chi^{\mathrm{eq}} - \chi^{\mathrm{eq}\top}\|_F / \|\chi^{\mathrm{eq}} + \chi^{\mathrm{eq}\top}\|_F$, which is zero iff the game is potential, at every λ [14]; (3) the *exact* σ_{EP} of the joint-profile chain; and (4) trajectory estimators of σ_{EP} that see only an observed path (Section 4). Families with α prescribed exactly are generated by unit-normalising the potential component of one random game and the harmonic component of another and mixing $(1-\alpha)\hat{P} + \alpha\hat{H}$; all sweeps below use 100 games per α level at seed 20260808 (regeneration: `make reproduce`).

Positioning. The behaviour of learning dynamics between the potential and harmonic extremes is an acknowledged obstacle [10]. Our angle is complementary and, to our knowledge, unoccupied: treat the stationary state *thermodynamically* — currents, dissipation, response symmetry, uncertainty bounds — with instruments calibrated where the correct reading is provable.

2 The $\alpha \times \lambda$ phase map: criticality escape and the supercritical wedge

Naively $\rho(SB) \propto \lambda$, since $S = \lambda C$ with $C = \mathrm{diag}(\sigma) - \sigma\sigma^\top$ the choice covariance. The sweep (9 λ × 11 α, medians over 100 games per cell, 9,900 solves) says otherwise (Figure 1).

Finding 1 (Criticality escape; F-0006)**.** *On potential games ($\alpha = 0$) the median $\rho(SB)$ is* non-monotone *in λ: $0.07 \to 0.51$ (at $\lambda = 1.7$) $\to 10^{-3}$ (at $\lambda = 15$). As $\lambda \to \infty$ the principal QRE branch concentrates on a strict equilibrium, $C \to 0$ exponentially, and $S = \lambda C \to 0$ despite the λ factor: the system escapes criticality* through *equilibrium concentration. At $\alpha = 1$, mixed equilibria keep C bounded away from zero and ρ grows without bound (6.9 at $\lambda = 15$).*

Finding 2 (The supercritical wedge; F-0006)**.** *Between the extremes a wedge of supercritical games opens. The* median *game first crosses $\rho = 1$ at $\alpha = 0.5$ (for $\lambda \geq 8.5$); a 0.2 fraction of games crosses already at $\alpha = 0.4$; the frontier $\lambda_c(\alpha)$ descends monotonically ($\lambda_c \approx 3$ by $\alpha = 0.8$). Inside the wedge the resolvent is near-singular, so response* magnitudes *(χ^{eq}, and hence $\mathcal{R}$ levels) are unreliable there — the meters themselves flag it; symmetry statements survive as direction-only reads.*

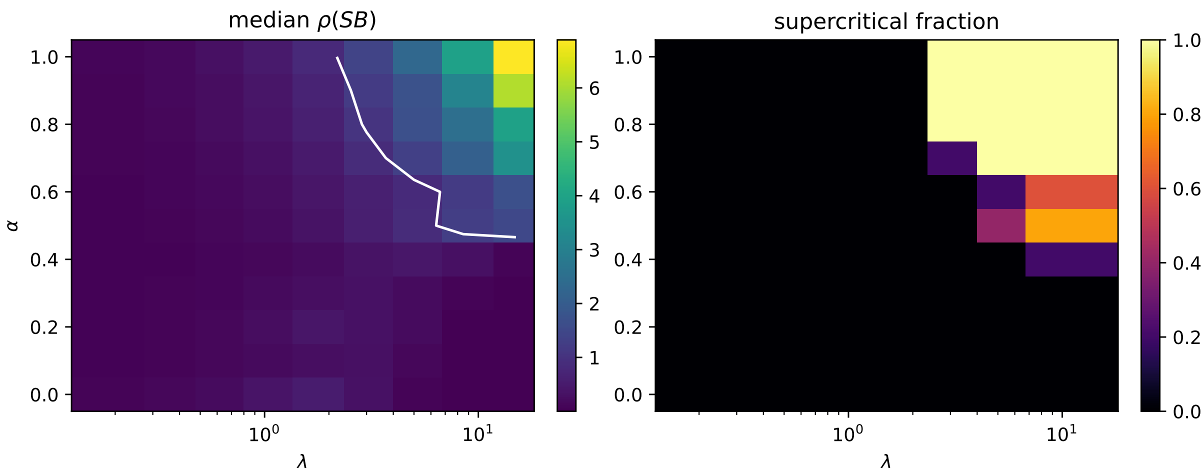

Figure 1: The $\alpha \times \lambda$ plane. Left: median $\rho(SB)$ with the $\rho = 1$ contour in white — non-monotone along the bottom edge (criticality escape), unbounded along the top. Right: fraction of games per cell with $\rho \geq 1$ — the supercritical wedge, frontier descending in α. Artifact: `phase_map_surface.json`.

Two consistency checks the map passes: the escape mechanism predicts no Hopf bifurcation of logit dynamics in full potential games ($B = B^\top$ makes the SB spectrum real; verified exactly across the λ grid), consistent with cycles requiring non-potential structure [11]; and the $\alpha = 0$ row of every dissipation meter reads zero to machine precision at all λ, as it must.

Both questions this section left open in v0.1 are now closed (unit `science.frontier`). The peak location is a *pure* scale fold: $\sigma(\lambda, s \cdot u) = \sigma(s\lambda, u)$ is an algebraic identity (checked numerically to error 0.0), so ρ depends on λ and the payoff scale only through their product and the peak is physical only in the normalised coordinate. And $\lambda_c(\alpha)$ — defined as the verified-unique crossing of the fixed-family median-ρ curve, with single-crossing checked at every level before bisection — is a clean monotone-descending frontier: 7.8 at $\alpha = 0.55$ down to 3.0 at $\alpha = 0.80$, with no median crossing below $\alpha \approx 0.5$ by $\lambda = 15$ at 40-game sampling (the earlier 5-game map's onset at $\alpha = 0.5$ is sampling variability of the median, recorded as such).

3 Response asymmetry and dissipation are distinct observables

Both $\mathcal{R}$ and σ_{EP} vanish exactly on potential games and read loudly on harmonic ones, so a natural conjecture (C1 in the programme's ledger) held that the four meters co-move tightly along the whole family. Marginally they do: over 1,000 games (100×10 α levels, $\lambda = 1.2$), Spearman $\rho(\sigma_{\text{EP}}, \mathcal{R}) = 0.993$.

Finding 3 (Conditional decoupling and sign reversal; F-0004)**.** *The marginal agreement is α-confounding. Conditional on the level, $\rho(\sigma_{\text{EP}}, \mathcal{R})$ is $+0.80..+0.88$ for $\alpha \leq 0.65$, degrades through $+0.61$ $(\alpha = 0.75)$ and $+0.32$ $(\alpha = 0.85)$, and* reverses *to -0.355 at $\alpha = 0.95$: among near-pure-harmonic games, larger response asymmetry associates with* less *dissipation (Figure 2). This finding originated as a red-team objection to our own headline correlation; the stratified metric is now a permanent artifact field.*

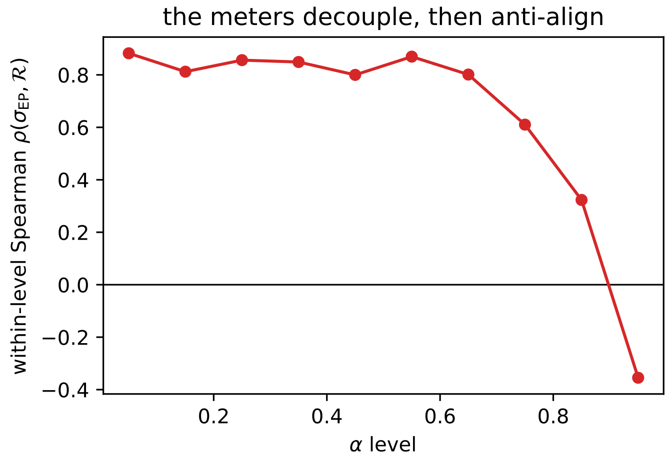

Figure 2: Within-level rank correlation between σ_{EP} and $\mathcal{R}$ along α. Strong coupling while a potential component modulates both meters; decoupling and sign reversal as it vanishes. Artifact: `chain_comovement.json`.

Mechanism, by preregistered test (F-0007). We proposed a repair: perhaps only $\mathcal{R}$'s *denominator* $\|\chi^{\mathrm{eq}} + \chi^{\mathrm{eq}\top}\|$ misbehaves at high α (H2), while the antisymmetric numerator keeps tracking dissipation (H1) — in which case a numerator-only meter would restore coupling. On the identical seeded families, paired by game: **H2 confirmed** — at $\alpha = 0.95$, $\rho(\mathcal{R}, 1/\|\chi^{\mathrm{eq}}+\chi^{\mathrm{eq}\top}\|) = 0.993$, rising monotonically from ≈ 0 at low α; $\mathcal{R}$'s within-level variation in the near-harmonic regime is entirely the shrinking symmetric response. **H1 refuted** — the numerator *also* decouples: $\rho(\|\chi^{\mathrm{eq}}-\chi^{\mathrm{eq}\top}\|, \sigma_{\mathrm{EP}}) = -0.37$ at $\alpha = 0.95$, down from $+0.77..0.86$ at $\alpha \leq 0.65$. No renormalisation of the response matrix recovers the dissipation ordering.

The structural reading. This negative result is the better result. χ^{eq} is a *local derivative* of the equilibrium map at the QRE point; σ_{EP} is a *global functional of the stationary flux* of the joint-profile chain. They co-vary exactly as long as a potential component modulates both, and part company when it vanishes. Consequently the instrument-scope table of [14] is not a convenience but a necessity: $\mathcal{R}$ answers "is this system potential?" (exactly, at every λ); σ_{EP} answers "how hard is it driven?"; and in the near-harmonic regime neither substitutes for the other. Conjecture C1 is thereby resolved as: co-movement for $\alpha \lesssim 0.65$, structural failure above. The crossover's λ- and size-dependence (open in v0.1) is now measured (F-0010): the *collapse* of within-level coupling at $\alpha = 0.95$ is universal — from ≈ 0.90 at $\alpha \leq 0.65$ to ≤ 0.25 in every condition tested ($\lambda \in \{0.8, 1.2, 2.0\}$, both 3×3 and 4×4) — while the *negative sign* is a second, λ-dependent effect ($+0.25 \to +0.03 \to -0.23$ across the three λ; -0.26 at 4×4), with each individual sign within $\sim$2 null-SD at $n = 60$. We record the methodological event plainly: the initial criterion (reversal in $\geq 3/4$ conditions) *failed* and the failure is the finding — the robust fact is decorrelation, not anti-alignment, and whether the sign tracks depth into the supercritical wedge is the sharp remaining question.

4 Dissipation from trajectories alone

Everything above used the generator. Data never provides it: only an observed sequence of profiles. Two estimators bridge the gap, each validated against the exact Schnakenberg meter on synthetic Glauber trajectories *before* any data contact (unit `thermo.estimators`; red-team signed off after one blocking objection, disclosed below).

Sampling identity. We simulate by uniformisation: the skeleton chain $P = I + L/\Lambda$ stepped with $\mathrm{Exp}(\Lambda)$ holding times. Two exact facts make the skeleton the right estimation object: it shares the chain's stationary law, and its per-step entropy production equals $\sigma_{\mathrm{EP}}/\Lambda$; moreover the $(k+1)$-block KLD of a stationary Markov chain equals k times the per-step entropy production, so the estimator below has a sharp target at every k.

KLD k-th order Markov [7]. The plug-in $\hat{\sigma}^{(k)} = \frac{1}{k\tau}\sum P(Y_{0:k})\log P(Y_{0:k})/P(Y_{k:0})$ recovers the exact meter with Spearman $\rho = 1.0$ and $\sim 1\%$ per-level error across the ten-α family (Figure 3, left), and reads $< 5\times10^{-3}$ on exact potential games (pure plug-in bias; truth is 0). It is data-hungry: the identity is a population statement, and with $n \ggg n_{\mathrm{states}}^{k+1}$ the estimator *underestimates* ($k = 4$ reads $\sim 17\%$ low at our validation sample size; the failure mode is itself a regression test).

TUR lower bound [5, 6]. From the first two cumulants of a time-integrated current J_T: $\sigma_{\mathrm{EP}} \geq 2\langle J_T\rangle^2/(\mathrm{Var}(J_T)\,T)$. A certified bound rather than a point estimate — a poor current choice loosens but never breaks it — which makes it the designated headline number for partially observed data [8]. Three methodological facts were learned adversarially and are now tests: (i) the bound is a *fixed-horizon* statement — fixing the jump count instead suppresses the Poisson timing fluctuation and inflated the "bound" 1.5× above the true EPR before we caught it; (ii) at M windows the sample version carries $+\mathrm{Var}/M$ and Jensen-on-$1/\widehat{\mathrm{Var}}$ biases, both now corrected; (iii) even debiased, the *point* estimate legitimately straddles the truth near equilibrium, where the TUR saturates — it exceeded the exact EPR at 2 of 10 sweep levels (max ratio 1.073). The certified statement is the lower bootstrap quantile, which stayed below the exact EPR at every level. The initial artifact language called the point estimate a certified bound; the red-team blocked the gate until the distinction was made structural (a separate `tur_epr_bound_ci` API) — we record this because the correction is part of the result.

Tightness is a diagnostic. The bound/exact ratio is ≈ 0.97 at $\alpha = 0.05$ (linear-response saturation) and drifts to 0.6–0.7 by $\alpha = 0.95$ (Figure 3, right): the TUR is tightest exactly where dissipation is smallest, i.e. it loses efficiency precisely where there is more to measure — the right failure direction for a conservative instrument.

5 Driving the system: quench thermodynamics

The readings so far are stationary. Real strategic environments are *driven*: incentive sharpness moves — information arrives, stakes change, λ ramps. Unit `thermo.protocols` instantiates the Hatano–Sasa decomposition [12, 13] exactly on the Glauber chain: for any distribution p

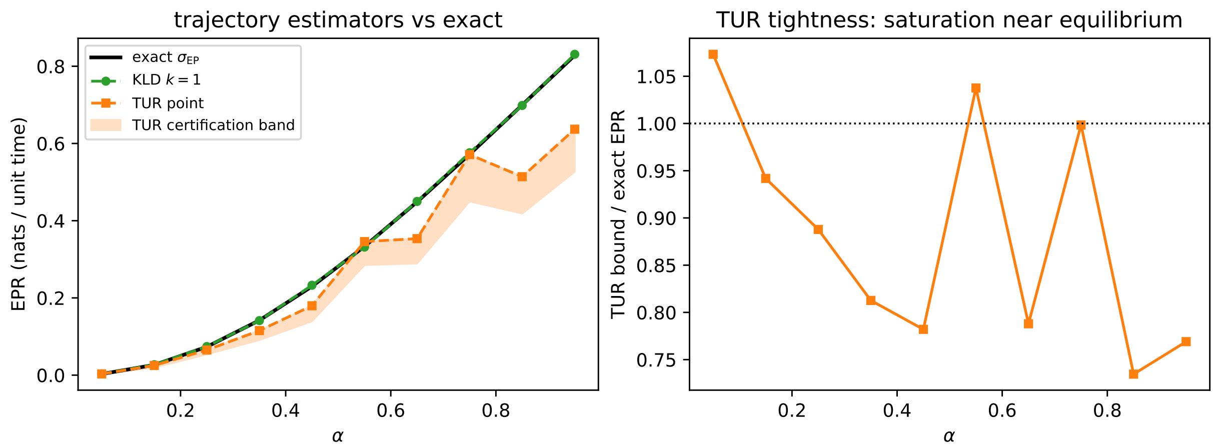

Figure 3: Left: exact EPR vs. its trajectory estimators along α ($\lambda = 1.5$; KLD from 8×60k stationary steps, TUR from 128 fixed-horizon windows). Right: TUR tightness — the bound saturates near equilibrium and degrades gracefully (and conservatively) as the drive grows. Artifact: `estimator_alpha_sweep.json`.

evolving under the generator at the current λ,

$$\sigma_{\rm tot}(p) \;=\; \underbrace{\sigma_{\rm hk}(p)}_{\text{holds the NESS}} \;+\; \underbrace{\sigma_{\rm ex}(p)}_{-\frac{d}{dt}D(p\,\|\,\pi)} \;,$$

both parts nonnegative, with $\sigma_{\rm hk} \equiv 0$ precisely under detailed balance — on potential games, dissipation is pure relaxation. For stepwise λ-protocols the Hatano–Sasa integral fluctuation theorem $\langle e^{-Y}\rangle = 1$ holds for *every* game, including off detailed balance where Jarzynski does not apply. One honesty note our red-team insisted on and we verified: for stepwise protocols the weighted-transfer computation of $\langle e^{-Y}\rangle$ telescopes to 1 algebraically, so it is an identity verification, not a bug-catcher; the correctness test is the independent sampled-path check, powered at $\alpha = 0.5$ where $\langle Y\rangle \sim 10^{-2}$ (CI $[0.995, 1.003] \ni 1$; artifacts `protocol_ift_checks.json`).

The driving-cost inversion (F-0012). A pre-registered scan (predictions written in config before the run; the intended prior commit aborted on a hook failure and the audit gap is recorded, not smoothed over) ramped $\lambda : 0.5 \to 3$ across the α family. Excess dissipation obeys the discrete quasi-static law $\langle Y\rangle \propto 1/K$ in the number of steps (measured slopes -1.02, -0.94) and *collapses* three orders of magnitude with α ($3.6 \times 10^{-2} \to 2.8 \times 10^{-5}$ nats), while accumulated housekeeping grows to 12.4 nats, exactly linear in protocol duration ($\sigma_{\rm hk}$ depends only on the generator and its π, so per-hold constancy is structural; the informative number is the burn *rate*, 0.77 nats per unit time at $\alpha = 0.95$). The cost of driving inverts across the family: potential games pay per *change* and drive for free in the fine-step limit; near-harmonic games pay *rent per unit time* whether driven or not.

The mechanism, found by the machine (F-0013). Why does excess collapse? Four quantitative hypotheses — a payoff-scale fold, the NESS-sensitivity floor $F = \sum_k D(\pi_{k-1}\|\pi_k)$, a spectral-gap lag model, and a quadratic strawman — were handed to an expected-free-energy experiment selector (unit `estimate.bayes`): each hypothesis predicts every candidate probe's outcome, the next probe maximises the BALD mutual information between hypothesis and outcome, and beliefs update by Bayes. The campaign chose the single most discriminating probe available ($\alpha = 0.95$, 1.19 nats of expected information) and resolved in one round: the *floor* hypothesis at posterior 0.996, then validated on all 19 unconsumed probes (median residual 0.002 dex). Scope, stated plainly: in the well-relaxed regime the floor formula is exact-in-the-limit ($\langle Y \rangle \to F$), so the campaign's content there is regime identification plus generalisation (the 3×3 family agrees to 0.1%); at fast quenches lag dominates and the floor formula fails by design. A second campaign (F-0014) closed that remainder at first order — with its own instructive failure first: the selector stopped confidently after one probe and the pre-registered held-out guard *refused* the verdict (`winner_failed_validation`), a lesson now institutionalised as the loop's `min_probes` stopping gate. The validated picture: excess is bracketed by the path-length floor ($\tau \to \infty$) and the frozen divergence $D(\pi_{\text{start}} \| \pi_{\text{end}})$ — exact *at* $\tau = 0$ but with a non-uniform approach — and a single spectral-gap crossover $\langle Y \rangle \approx F + (D_{\text{frozen}} - F)e^{-g\tau}$ holds to ~ 0.1 dex on monotone NESS paths (held-out median 0.089). Where the NESS path is loop-like ($\pi_{\text{end}} \approx \pi_{\text{start}}$ with excursions between, as at $\alpha = 0.95$ with a long ramp) the formula fails by ~ 1 dex, and that failure is reported, not averaged away. A third campaign (F-0015) then replaced the global formula with a *per-step, path-aware* recursion — $p_k = \pi_k + (p_{k-1} - \pi_k)e^{-g_k\tau}$ with g_k the gap at the current λ_k, Y accumulated along the tracked distribution — which dominates on both median (0.076 vs 0.13 dex, 3×3 games included) and worst case (0.44 vs 1.11: the loop-path failure is structural to endpoint comparison and vanishes when the path is followed). Two process notes belong in print: the campaign's first two-mode "result" was retracted as an implementation artifact caught by adversarial review, and the corrected two-mode model *still* loses to one mode — truncating the fast remainder at each switch errs more than damping the whole deviation at the gap rate.

6 First data contact: a power market

The estimators of Section 4 were built synthetic-first; their first real-data application (unit `domains.electricity`) is CAISO's SP15 hub — day-ahead hourly and real-time 5-minute locational marginal prices, July 2026. The sequence of nulls it took to read them honestly is the methods contribution; the reading itself is the empirical one.

Three structural lessons, each now a permanent guard. (i) Price-*value* discretization is provably blind to loop irreversibility — a periodic drive retraces the same 1-D path, so its reversal visits identical value transitions (unit-tested on a noisy sine; then observed on the real data to four decimals: the value-space reading sits exactly on its null). The loop becomes visible only in the phase embedding (price bin, Δ-sign). (ii) A plain shuffle null fails twice (destroys persistence; the embedding of an i.i.d. series is structurally asymmetric), and a plain FT surrogate fails on heavy tails (LMPs carry excess kurtosis ≈ 130; Gaussianisation distorts the sign-flip rate). Amplitude-adjusted surrogates repair the marginal but still cannot bracket the data's direction-persistence — the reading escapes the null band on the *low* side, which is neither detection nor certified null. (iii) The null that decides is the *reversibilized Markov* surrogate: fit the embedded pair counts, symmetrize the flux, simulate the detailed-balance chain with identical persistence. It satisfies detailed balance exactly, and its false-positive rate on truly reversible data matches α.

The reading (F-0009, adversarially verified). Against the persistence-matched null, the day-ahead hourly series *violates pair-level detailed balance*: 0.0447 nats/hour vs. null $q_{99} = 0.0291$, empirical $p < 0.01$, robust to surrogate seeds, binning (2/3/4 price bins), tie-breaking, Bonferroni across markets and null classes, and reversible second-order structure. Weekly stratification shows the signal has consistent sign in all five weeks but concentrates in the high-ramp second half (per-week ratios $1.6, 1.5, 3.0, 9.1, 4.2$; the detection is statistically driven by the scarcity weeks, stated wherever the number is used). In physical terms: the diurnal loop — night valley, morning ramp, evening peak, decline — dissipates ≈ 1.1 nats per day at hourly resolution. The real-time 5-minute series, by contrast, is genuinely at-null at $k \in \{1, 2\}$ in the matched class: at that sampling rate dwell dominates the pair statistic. And the domain's stylised two-generator auction model, asked to rationalise the observed price dispersion, is *rejected* by it (the model's dispersion ceiling sits 40% below the spike-driven observation) — the conditional-λ pipeline reports the rejection rather than a number.

One retraction is part of this record: an intermediate "certified null" claim (v0.1 of finding F-0008) was withdrawn on adversarial review and replaced by the scoped verdicts above. The audit trail — four null classes, two red-team reviews, the retraction — ships with the repository.

7 Discussion

Three findings, one arc. The phase map says *where* the interesting physics lives: not at the calibration anchors but in the wedge between them, where strategic feedback goes supercritical and response magnitudes stop being trustworthy. The decoupling result says *what* lives there: the response layer and the dissipation layer, interchangeable to the eye along the marginal sweep, are structurally different observables — a local derivative and a global flux — and the near-harmonic regime is where the difference becomes measurement-relevant. The estimator result says the dissipation layer is *reachable from data*: rank-exact recovery when the chain is observed, and a certified lower bound that survives partial observation, with its tightness itself carrying regime information.

What would change our minds. The wedge frontier could dissolve under payoff-scale folding (the $\alpha = 0$ peak at $\lambda \approx 1.7$ is suspiciously close to scale/2); the decoupling crossover at $\alpha \approx 0.7$ could move materially with λ or with action-space size, which would demote it from structural fact to small-game artefact; and the suspicious coincidence that F-0004's decoupling onset sits *inside* the supercritical wedge deserves either a mechanism or a debunking. All three are queued as finer sweeps with committed seeds.

Toward data. The empirical programme (Stage 3 of the repository) points these meters at systems whose α nobody knows: retail pricing panels and electricity bidding. The division of labour established here is what makes that honest: the reciprocity test for potentiality (exact, λ-free as a symmetry statement), the TUR bound for dissipation (certified, partial-observation-tolerant), and neither doing the other's job. The nearest empirical precedent is the entropy-production detection of an Edgeworth price cycle in experimental data [9]; the instruments here are built to make such readings routine, calibrated, and falsifiable.

Provenance

Every quantitative claim in this draft maps to a committed, gate-checked artifact regenerated from fixed seeds by `make reproduce`:

Claim	Artifact	G
Phase surface, wedge, criticality escape	`phase_map_surface.json`	s
Marginal and within-level $\rho(\sigma_{\mathrm{EP}}, \mathcal{R})$	`chain_comovement.json`	d
H1/H2 mechanism test	`decoupling_mechanism.json`	s
KLD/TUR ground truth (named games)	`estimator_ground_truth.json`	t
KLD $\rho = 1.0$; TUR certification per level	`estimator_alpha_sweep.json`	t
Zero calibration (potential games, real network)	`equilibrium_reads_zero.json`, `sioux_falls_calibration.json`	d

Each gate closed only after adversarial review; the objections and their dispositions (including the blocking objection to the TUR certification language, and the red-team origin of the decoupling finding itself) are in `gates/*.yaml` in the repository. Claims carry tiers in `memory/claims.md`; conjecture C1's partial falsification is recorded there, not smoothed over.

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